Hochschule Fulda
University of Applied Sciences

GLOBAL DISTRIBUTED SOFTWARE DEVELOPMENT

AI5088

Master Team Project Summer 2026 - Revised Milestone 1 - Team 2

SkillBridge

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Milestone 1 - Revised Version

1. Executive Summary

In today's academic environment, many students need extra support outside regular lectures, but finding the right tutor can be difficult, time-consuming, and expensive. Students often do not know where to look for reliable help, while tutors may struggle to reach students and organise their teaching activities. SkillBridge is planned as a digital tutoring platform that connects students with tutors and their learning support services in one simple application. The goal of the project is to make academic help easier to find, easier to organise, and more accessible for different learning needs.

SkillBridge's unique selling point is that it brings tutor discovery, session booking, communication, learning resources, and student progress tracking into one tutor-focused platform.Students can discover tutors and learning content based on subject, availability, rating, price, and type of support. They can search, sort, and filter available tutors or content, book one-on-one tutoring sessions, communicate with tutors through in-site chat, and view their learning activity in a personal dashboard. Tutors can create profiles, describe their services, set available timeslots, and share learning materials such as notes, slides, videos, links, or exercises. The platform is designed for users with different levels of technical experience, so both students and tutors should be able to use it easily without needing advanced technical knowledge.

The planned advantage of SkillBridge is that it focuses specifically on tutoring and personalised academic support, rather than only general classroom or course management. Existing platforms such as Google Classroom, Moodle, Canvas LMS, and Schoology are mainly designed for managing courses, sharing materials, and supporting classroom communication. SkillBridge adds a more service-focused approach by combining tutor discovery, session booking, tutor-student communication, content sharing, student dashboards, and admin-supported content review in one place. AI-supported content moderation is also planned to help admins identify inappropriate, misleading, spam-like, or copyright-sensitive content before it becomes visible to users. Since this is a university software engineering project, payment-related features will only be considered at a demonstration or planning level and will not represent a real payment system.

Our student startup team, Team 2, consists of Anish Bade, Hrishav Banik, Subham Gaurav, and Nitya Lakhamanbhai Vadhel. The team combines backend development, frontend development, GitHub management, teamwork, and software design responsibilities. By working together, we aim to build a practical, user-friendly, and scalable prototype that addresses a real student problem: finding reliable academic support in a simple and organised way. SkillBridge is therefore not only a tutoring application but also a learning support platform designed to make education more connected, flexible, and accessible.

2. Personas and Use Cases

2.1 Personas

Persona 1: Super Admin

The Super Admin is responsible for overseeing the entire platform across user groups. Their primary goal is to ensure the system operates smoothly, securely, and consistently for all users. They manage high-level settings, monitor platform performance, and oversee other platform admins.
Super Admins often face challenges related to maintaining system-wide consistency, handling large numbers of users, and responding to technical or operational issues quickly. Since they support many stakeholders, efficiency and centralized control are very important to them.

Persona 2: Admin

The admin manages the daily operations of the platform. Their goals include approval of new users(students and tutors), moderation of content, reviewing tutor profiles, monitoring content, and ensuring each user has access to the resources they need.
Admins often work under time constraints and may handle multiple responsibilities at once. Common pain points include managing large amounts of content and ensuring every bit of information on the platform is accepted as per policies.

Persona 3: Tutor

The Tutor is an educator or instructor who uses the platform to guide and support students in their learning activities through different kinds of tutoring services. Tutors typically have varying categories of technical expertise, but are mostly focused on teaching rather than system management. Their main goals are to deliver learning materials, track earnings, provide feedback, and chat with students.
Tutors may struggle with managing multiple appointments and one-on-one sessions. They also need efficient ways to organise teaching activities without spending excessive time on managing the platform. A clean and user-friendly environment is important so they can focus on educational outcomes rather than technical complexity.

Persona 4: Student

The Student is the primary learner using the platform to access educational content and communicate with tutors through chat. Students may range from beginners with limited technical skills to highly experienced digital users. Their goals are to request appointments, learn effectively, stay organised, track their progress, and receive timely support from tutors.
Students often experience challenges related to keeping up with deadlines, finding relevant learning materials, maintaining motivation in online or digital learning environments and most importantly, failure in finding proper advisors for their studies. They benefit from systems that are simple to navigate, provide clear guidance, and make communication with tutors straightforward. Accessibility, convenience, and engagement are key factors for this persona.

Persona 5: Guest User

The Guest User is a visitor who can explore SkillBridge before creating an account. Their goal is to understand what the platform offers and see whether it can help them find suitable academic support. Guest users can search for selected tutors, subjects, or learning support categories, but they cannot access all details or use full platform features such as booking sessions, messaging tutors, saving favourites, or viewing complete learning resources.

Guest users may be unsure whether they want to join the platform, so the system should give them enough preview information to attract their interest and encourage registration. For example, they may see limited tutor profiles, subject categories, ratings, and examples of available support. Content shown to guest users should only include approved and safe public information, supported by admin review and AI-supported moderation. This helps SkillBridge promote trust while encouraging students to register with their Hochschule Fulda email to access the full platform.

2.2 Use Cases

UC-1: Tutor Registration with Offered Services and Timeslots

- **Actor:** Tutor
- **Goal:**
  The tutor wants to join SkillBridge, present their tutoring services, and make their available timeslots visible to students.
- **Description:**
  A tutor joins SkillBridge because they want to offer academic support to students at Hochschule Fulda. The tutor may have different levels of technical experience, so the platform should provide a simple and clear onboarding experience. By completing this use case, the tutor becomes part of the platform and can offer one-on-one or group tutoring services to students who need academic help.

UC- 2: Student Registration and Login

- **Actor:** Student
- **Goal:**
  The student wants to join SkillBridge and access tutoring services, learning content, and communication features.
- **Description:**
  A student uses SkillBridge because they need reliable academic support from tutors within the Hochschule Fulda community. The student may be a beginner or experienced digital user, so the registration and login experience should be easy to understand. After joining the platform, the

student can search for tutors, view their learning content, book tutoring sessions, and communicate with tutors in one organised place.

UC-3: Student Searches, Sorts, and Filters Tutors or Content

- **Actor:** Student
- **Goal:**
  The student wants to find a suitable tutor or learning material based on their academic needs and personal preferences.
- **Description:**
  A student visits SkillBridge to find support for a specific subject, course, or academic problem. The student may not know which tutor or material is best at first, so they need a simple way to explore available options. This use case helps students discover tutors or content based on factors such as subject, price, availability, rating, popularity, or content type, so they can choose the most relevant academic support.

UC-4: Student Books a One-on-One Tutoring Timeslot

- **Actor:** Student
- **Goal:**
  The student wants to arrange a private tutoring session with a tutor who matches their learning needs.
- **Description:**
  A student finds a tutor or service provider who matches their learning or personal development needs and wants one-on-one guidance or support. The student may need help with academic subjects, musical instruments, sports training, language learning, creative skills, or other specialised services. Through this use case, the student books a suitable one-on-one session, while the tutor or service provider can keep track of upcoming appointments and scheduled sessions.

UC-5: AI-Supported Content Moderation

- **Actor:** Admin, supported by the system
- **Goal:**
  The admin wants to maintain a safe, trusted, and policy-compliant learning environment.
- **Description:**
  SkillBridge supports safe and trusted learning by checking uploaded content, tutor materials, descriptions, and shared media before they become visible to students. This use case mainly supports admins, who may need to review many content submissions and identify inappropriate, misleading, spam-like or copyright-sensitive material. The purpose is to support platform quality and help students and tutors use SkillBridge with confidence.

UC - 6 Super Admin Manages Admins and Platform Settings

- **Actor:** Super Admin
- **Goal:**
  The super admin wants to manage platform-level control, admin accounts and general system settings.
- **Description:**
  The super admin uses SkillBridge to keep the overall platform organised for all user groups. This actor is usually more experienced and responsible for high-level control, such as managing admin users, monitoring platform activity and maintaining general platform rules. This use case ensures that SkillBridge can be managed properly as the number of students, tutors, content items, and administrative tasks grows.

3. List of Main Data Items and Entities

Below is a table defining key data items and entities used in the application, including columns for Index, Entity/Item Name, and Description (Purpose/Usage).

Index	Entity / Data Item	Description / Purpose
1	User	A general person using SkillBridge. A user can be a student, tutor, admin or super admin.
2	User Role	Defines the user type and privileges, such as Student, Tutor, Admin or Super Admin.
3	Student Profile	Represents student information used for finding tutors, booking sessions, viewing content and tracking activity.
4	Tutor Profile	Represents tutor information such as subjects, skills, teaching style, availability, pricing and approval status.
5	Subject / Category	Represents the academic or skill area connected to tutoring, such as programming, mathematics, language learning or exam preparation.
6	Tutor Service	Describes the type of support offered by a tutor, such as one-on-one tutoring, group tutoring or learning material support.
7	Timeslot / Availability	Represents the time periods when a tutor is available for tutoring sessions.
8	Booking / Session	Represents a planned tutoring appointment between a student and a tutor.

9	Learning Content	Represents study materials shared by tutors, such as notes, slides, videos, links or exercises.
10	Content Review / Moderation	Represents the approval, rejection, flagging or AI-supported review of tutor profiles, uploaded content or reported items.

4. Initial List of Functional Requirements

1. Fulda-only registration verification

The system shall allow only users with an email address ending in hs-fulda.de to register and access the platform as students or tutors. This is the basic affiliation gate that enforces the exclusive community requirement.

2. Role-based account creation

The system shall support at least three roles: student, tutor, and administrator. A user's role shall determine which features and pages they can access.

3. Tutor profile creation

The system shall allow approved tutors to create and edit a detailed profile, including name, subjects, academic program, semester, short bio, teaching style, languages, and availability. This gives students enough information to compare tutors before contacting them.

4. Media-rich tutor profile uploads

The system shall allow tutors to upload supporting media such as CV files, profile images, short videos, audio introductions, and optional sample materials. This makes profiles more personal and more informative than a plain text listing.

5. Tutor profile approval workflow

The system shall require every new or edited tutor profile submission to be approved by an administrator before it becomes publicly visible. This protects the platform from inappropriate or low-quality content.

6. Tutor search

The system shall provide a search function that allows students to find tutors by subject, keywords, language, availability, rating, price and format.

7. Advanced filtering and sorting

The system shall allow students to sort tutors by subject, relevance, lowest price, highest rating, newest profile, and best availability match. This helps users narrow down the right tutor quickly.

8. Tutor review and rating system

The system shall allow students to rate and review tutors after an interaction or a completed tutoring session. Reviews shall be visible on tutor profiles to support trust and decision-making.

9. In-site messaging

The system shall provide secure in-site messaging(chat) between students and tutors for questions, scheduling, and follow-up communication. No external email client shall be used for user-to-user communication.

10. Tutor contact request

The system shall allow students to send a contact or booking request to a tutor directly from the tutor profile or search results. This reduces the effort needed to initiate tutoring.

11. Availability calendar

The system shall allow tutors to publish weekly availability and students to view available time windows before contacting them. This avoids back-and-forth messaging and makes scheduling easier.

12. Session request management

The system shall allow tutors to accept, decline, or propose a new time for student requests. The platform shall notify both parties about request status changes.

13. Study-trial recommendation

The system shall suggest tutors based on the student's faculty, semester, previously searched subjects, and previously linked prerequisite topics. This is a unique feature that makes the platform feel more intelligent and personalised.

14. "Exam rescue" mode

The system shall let students mark tutoring needs as urgent, such as exam preparation, assignment deadlines, or last-minute help. The platform shall then highlight tutors who have the nearest matching availability.

15. Resource sharing on tutor profiles

The system shall allow tutors to attach study resources such as notes, example solutions, links, or preparation checklists to their profiles or course profiles. This adds extra value beyond simple contact information.

16. Subject tags and skill badges

The system shall allow tutors to add standardised subject tags and badges such as “exam prep,” “bilingual support,” “first-year support”, or “lab help.” These tags improve search quality and help tutors stand out.

17. User report feature

The system shall allow students and tutors to report suspicious, offensive, misleading or spam content. Reported items shall be queued for administrator review.

18. Favourites or shortlist

The system shall allow students to save preferred tutors to a shortlist for later comparison. This is helpful when students want to review several candidates before deciding.

19. Compare tutors

The system shall allow students to compare selected tutors side by side using key attributes such as price, rating, subjects, languages, availability and media content. This improves decision quality.

20. Mobile-responsive interface

The system shall render correctly on desktop and mobile screens, including search, profile viewing, messaging, and review submission. The design should remain usable on phones even though the main target is desktop/laptop.

21. Media upload validation

The system shall check uploaded files for acceptable type, size, and basic content constraints before storing them. This prevents broken uploads and reduces misuse of the media features.

22. Notification system

The system shall notify users about new messages, contact requests, profile approvals, review submissions, and schedule updates inside the platform. Notifications can help users respond faster without needing email integration.

23. Tutor visibility controls

The system shall allow tutors to temporarily hide their profile, pause availability, or switch themselves to “not accepting requests.” This is useful during exams, holidays, or busy periods.

24. Administrative dashboard

The system shall provide an admin dashboard showing pending approvals, flagged content and active users. This helps administrators monitor the platform efficiently.

25. Search analytics for improvement

The system shall record aggregated search trends such as most requested subjects, peak tutoring times, and common unmet requests. These insights can guide future platform improvements and help the startup adapt the service to its clients' (students') needs.

26. Payment service flow

Students will pay for the service they are taking, but it will only be a dummy one.

27. Account and content deletion

The system shall allow administrators to deactivate or delete users, tutor profiles, and inappropriate content when required by policy or moderation decisions. This provides essential control over platform quality and safety.

5. Initial List of Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from the chosen cloud server
2. Application shall be optimised for standard desktop/laptop browsers, e.g. must render correctly on the two latest versions of the two major browsers
3. All or selected application functions must render well on mobile devices
4. Data shall be stored in the database on the team's deployment cloud server.
5. No more than 50 concurrent users shall be accessing the application at any time
6. Privacy of users shall be protected, and all privacy policies will be appropriately communicated to the users.
7. The language used shall be English (no localisation needed)
8. Application shall be very easy to use and intuitive
9. The application should follow established architecture patterns
10. Application code and its repository shall be easy to inspect and maintain
11. Google Analytics shall be used (optional for Fulda teams)
12. No e-mail clients shall be allowed.
13. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
14. Site security: basic best practices shall be applied (as covered in the class) for the main data items
15. Application shall be media-rich (images, video, etc.). Media formats shall be standard as used in the market today

16. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development
17. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set up by class instructors and started by each team during Milestone 0
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages: "Fulda University of Applied Sciences Software Engineering Project, Summer 2026, For Demonstration Only" at the top of the WWW page. (Important so as not to confuse this with a real application).

6. Competitive Analysis

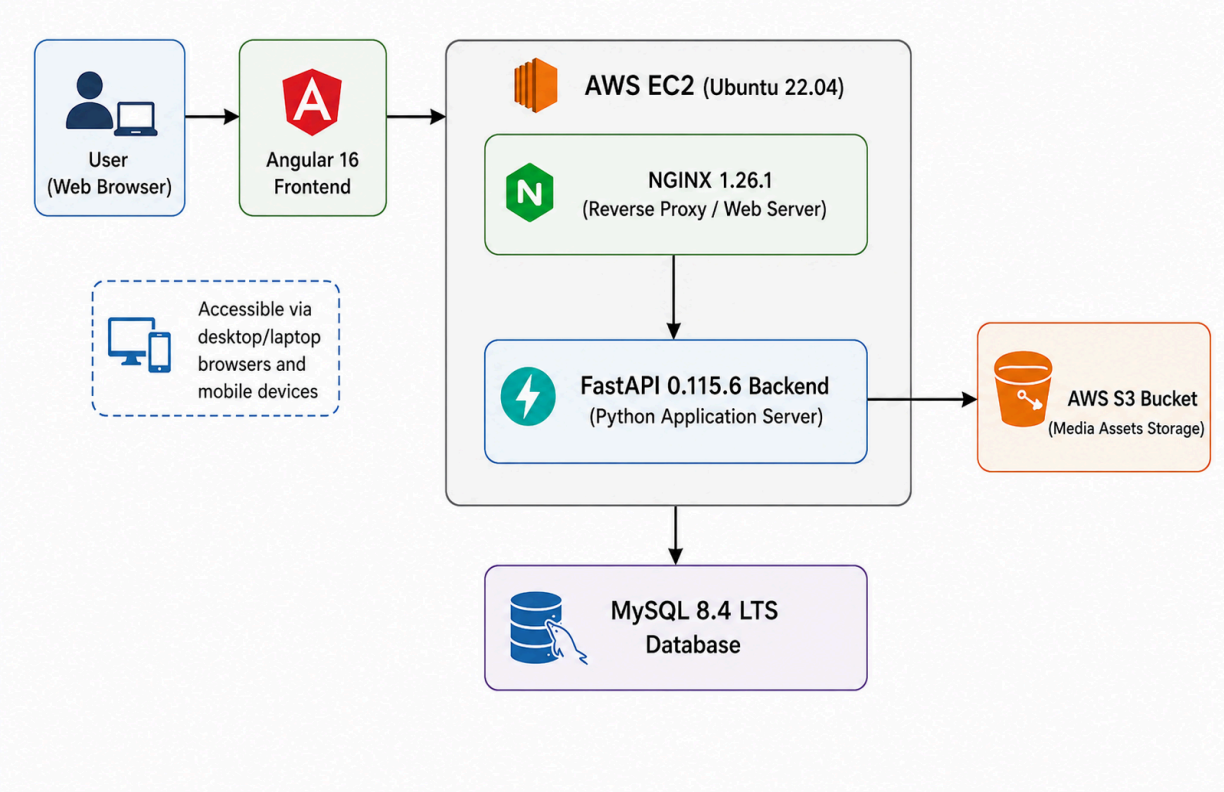
Features	***SkillBridge***	Google Classroom	Moodle	Canvas LMS	Schoology
Signup/Login (Exclusive to Hochschule Fulda email)	✔️	❌	❌	❌	❌
Search, Sort and Filter(Content or tutor)	✔️	Limited	Limited	✔️	✔️
Content Screening and Admin approval	✔️	❌	❌	Limited	Limited
Student activity dashboard	✔️	❌	❌	✔️	✔️

Chat Support between tutor and student	✔️	Limited	❌	✔️	❌
Pay-per-use tutoring/content	✔️	❌	❌	❌	❌

SkillBridge is planned as a tutoring-focused platform rather than a general learning management system. Existing platforms such as Google Classroom, Moodle, Canvas LMS and Schoology mainly support course management, learning material sharing and classroom communication. SkillBridge's planned advantage is that it combines tutor discovery, tutor filtering, one-on-one tutoring bookings, student dashboards, direct tutor-student communication, admin-supported content approval and pay-per-use tutoring or content access in one platform. This makes SkillBridge more focused on personalised academic support and flexible tutoring services, while the compared platforms are mostly designed for structured courses and classroom-based learning.

7. High-Level System Architecture and Technologies Used

- **Server Host:** Amazon Web Services EC2, 1 vCPU, 2 GB RAM
- **Operating System:** Ubuntu 22.04 LTS
- **Database:** MySQL 8.4 LTS
- **Web Server:** NGINX 1.26.1
- **Server-Side Language:** Python
- **Backend Web Framework:** FastAPI 0.115.6
- **Frontend Web Framework:** Angular 16
- **IDE:** Visual Studio Code
- **Supported Browsers:** Latest two versions of Google Chrome and Mozilla Firefox
- **Deployment Platform:** AWS EC2 on Ubuntu
- **Media storage:** AWS S3 bucket for images, videos, documents, and other media assets


User
(Web Browser)
Angular 16
Frontend
AWS EC2 (Ubuntu 22.04)
NGINX 1.26.1
(Reverse Proxy / Web Server)
Accessible via
desktop/laptop
browsers and
mobile devices
FastAPI 0.115.6 Backend
(Python Application Server)
AWS S3 Bucket
(Media Assets Storage)
MySQL 8.4 LTS
Database

8. Team and Roles

Name	**Email Address**	**Role**
Anish Bade	anish.bade@informatik.hs-fulda.de	Team Lead/ Backend Lead
Hrishav Banik	hrishav.banik@informatik.hs-fulda.de	Frontend Lead
Subham Gaurav	subham.gaurav@informatik.hs-fulda.de	GitHub Master
Nitya Vadhel	nitya-lakhamanbhai.vadhel@informatik.hs-fulda.de	Frontend Lead

9. Checklist

Checklist for team organisation with DONE/ ON TRACK/ ISSUE status.

Task Item	**Status**
Finalised meeting schedule	DONE
GitHub Master chosen	DONE
Agreed on Tech Stack	DONE
Learning approved tech stack	DONE
The team’s understanding of Milestone 1	DONE
GitHub repository organized	DONE